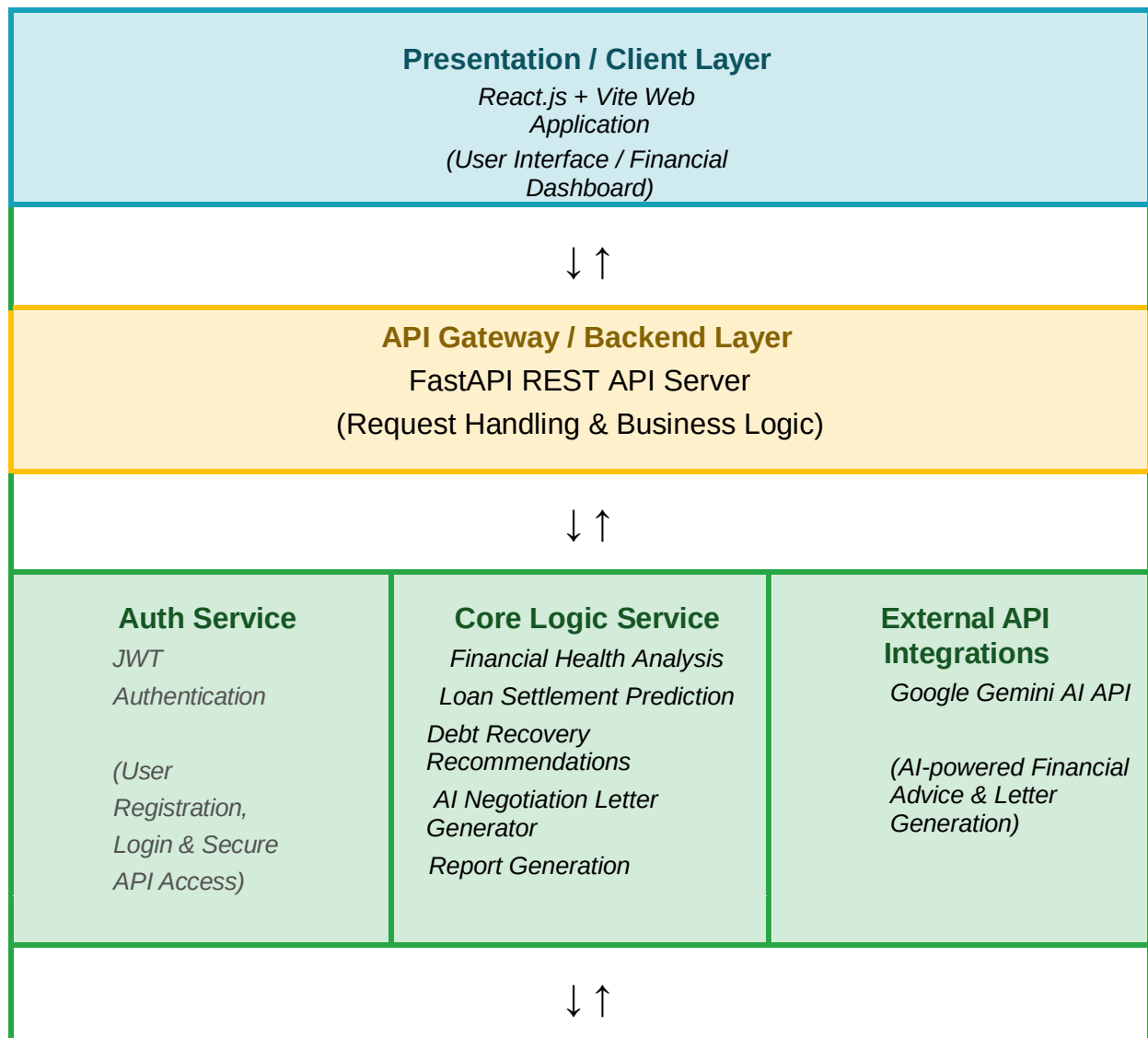


Solution Architecture

Date	28 June 2026
Team ID	xxxxxx
Project Name	FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Data / Storage Layer

SQLite Database
(SQLAlchemy ORM)

Tables:

- *Users*
- *Loans*
- *Financial Profiles*
- *Settlement Records*
- *AI Generated Reports*

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface for user registration, loan management, financial health dashboard, settlement prediction, AI-generated negotiation letters, and reports.	React.js, Vite, HTML, CSS, JavaScript
API Gateway	Handles incoming API requests, validates user data, processes business logic, and communicates with the database and AI services.	FastAPI, Uvicorn
Authentication Service	Authenticates users using secure JWT tokens and protects application endpoints.	JWT Authentication, FastAPI Security
Database / Storage	Stores user information, loan details, financial profiles, settlement history, AI-generated reports, and application data.	SQLite, SQLAlchemy ORM
External API Integration	Uses Google's Gemini AI to generate personalized financial advice, settlement strategies, and negotiation letters.	Google Gemini API
Testing Layer	Validates API endpoints, authentication, database operations, and application functionality.	PyTest